

Cybenko's Theorem (Universal Approximation Theorem)

Let:

$$K = [0, 1]^n$$

Define a single-hidden-layer network as follows:

$$N(x) = \sum_{j=1}^m \alpha_j \sigma(w_j^T x + b_j)$$

where:

- $x \in \mathbb{R}^n$
- $w_j \in \mathbb{R}^n$
- $b_j \in \mathbb{R}$
- $\alpha_j \in \mathbb{R}$
- $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ is the activation function.

For the sigmoid function:

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$

Cybenko's Theorem states that if σ is continuous and sigmoidal, meaning:

$$\lim_{t \rightarrow -\infty} \sigma(t) = 0$$

and

$$\lim_{t \rightarrow +\infty} \sigma(t) = 1$$

then the set

$$\mathcal{N}_\sigma = \left\{ \sum_{j=1}^m \alpha_j \sigma(w_j^T x + b_j) : m < \infty \right\}$$

is dense in the space

$$C([0, 1]^n)$$

That is:

$$\overline{\mathcal{N}_\sigma} = C([0, 1]^n)$$

And this is precisely the heart of the theorem.

What Does “Dense” Mean?

This part is very important.

When we say

$$\overline{\mathcal{N}_\sigma} = C(K)$$

we do not mean that the network is exactly equal to the function.

Rather, for every

$$f \in C(K)$$

and every

$$\varepsilon > 0$$

one can find a network such that:

$$\|f - N\|_\infty < \varepsilon$$

where:

$$\|f - N\|_\infty = \sup_{x \in K} |f(x) - N(x)|$$

So the network can get arbitrarily close to any continuous function, to any desired accuracy.